

5.5 — Design Human Review Workflows and Confidence Calibration

Concept

AI models rarely operate with uniform certainty. A model may confidently extract a company name from an invoice while being genuinely uncertain about an ambiguous date format. Human review workflows exist to catch these failures before they propagate — but reviewing everything is not scalable. Effective system design routes only the right items to human reviewers.

When to Involve Humans

Human review is warranted when:

- **Model confidence is low** — a confidence score falls below a calibrated threshold.
- **Stakes are high** — errors in medical records, financial figures, or legal documents carry outsized consequences.
- **Document type is novel** — the model has limited training signal for a new category.
- **Outputs affect downstream automation** — a mistake will be silently compounded by subsequent steps.

The goal is not to review everything, but to review the right things.

Field-Level Confidence Scores

Aggregate document-level confidence hides important variation. A model might be 97% confident about a vendor name extracted from a structured header but only 58% confident about a handwritten date in the footer. Surfacing confidence at the field level allows the system to flag only the uncertain fields for review, rather than discarding the entire document.

```
interface ExtractedField {
    value: string;
    confidence: number; // 0.0 – 1.0
    source: "structured" | "unstructured" | "inferred";
}

interface DocumentExtraction {
    documentId: string;
    documentType: string;
    fields: Record<string, ExtractedField>;
}
```

```
function flagFieldsForReview(
  extraction: DocumentExtraction,
  threshold: number
): string[] {
  return Object.entries(extraction.fields)
    .filter(([field]) => field.confidence < threshold)
    .map(([fieldName]) => fieldName);
}

// Example output: ["invoiceDate", "taxId"] – only these fields need review
const flagged = flagFieldsForReview(extraction, 0.8);
```

Stratified Sampling

Reviewing a representative subset — rather than every document — lets teams measure system quality without reviewer burnout. Stratified sampling ensures the sample covers all document types, confidence bands, and edge cases proportionally.

```
interface SampleConfig {
  documentType: string;
  confidenceBand: "high" | "medium" | "low";
  sampleRate: number; // fraction to review, e.g. 0.05 = 5%
}

function shouldSampleForReview(
  extraction: DocumentExtraction,
  configs: SampleConfig[]
): boolean {
  const overallConfidence = computeMeanConfidence(extraction);
  const band = classifyBand(overallConfidence);

  const config = configs.find(
    (c) =>
      c.documentType === extraction.documentType && c.confidenceBand === band
  );

  if (!config) return false;
  return Math.random() < config.sampleRate;
}
```

Higher sample rates for low-confidence bands and novel document types, lower rates for well-understood, high-confidence categories. This allocates reviewer capacity where it adds the most value.

Calibration: Confidence vs. Actual Accuracy

Model confidence scores are not inherently well-calibrated. A model that reports 90% confidence on a field may only be correct 70% of the time in practice — this is called overconfidence. Calibration is the process of

measuring the relationship between stated confidence and observed accuracy, then adjusting thresholds accordingly.

```
interface CalibrationBucket {
  confidenceRange: [number, number];
  totalPredictions: number;
  correctPredictions: number;
}

function computeAccuracy(bucket: CalibrationBucket): number {
  return bucket.correctPredictions / bucket.totalPredictions;
}

// After measuring: if 80–90% confidence bucket has only 65% actual accuracy,
// lower the review threshold for this document type to 0.90
function adjustThreshold(
  buckets: CalibrationBucket[],
  targetAccuracy: number
): number {
  const calibrated = buckets
    .filter((b) => computeAccuracy(b) >= targetAccuracy)
    .sort((a, b) => a.confidenceRange[0] - b.confidenceRange[0]);

  return calibrated.length > 0 ? calibrated[0].confidenceRange[0] : 0;
}
```

Run calibration per document type. A model calibrated on invoices may be systematically overconfident on purchase orders.

Review Queue Design

Review queues should be priority-ordered, not FIFO. Items with the lowest confidence scores and highest downstream impact should surface first.

- Assign a **review priority score** combining confidence, document type risk weight, and recency.
- Surface flagged fields inline rather than the full document to reduce cognitive load.
- Record reviewer corrections and feed them back into calibration measurements.

Common Mistakes

- **Single threshold for all document types.** Calibration differs by type. A single threshold will over-review high-confidence types and under-review low-confidence ones.
- **Conflating model confidence with accuracy.** A 90% confidence score is a model's internal estimate, not a guarantee. Always measure actual accuracy against ground truth.
- **Random sampling instead of stratified.** Pure random samples under-represent rare document types and edge cases that matter most.

- **Ignoring reviewer feedback.** Corrections made during review are ground-truth data. Failing to collect and act on them means calibration never improves.
- **Field-level confidence not surfaced to reviewers.** When reviewers see only a pass/fail flag, they re-verify the entire document instead of the uncertain fields — wasting time and introducing review fatigue.

References

- Anthropic Documentation — [Build with Claude: Human-in-the-loop patterns](#)
- Anthropic Documentation — [Tool use: structured outputs and validation](#)
- Niculescu-Mizil & Caruana (2005) — *Predicting Good Probabilities with Supervised Learning* (foundational calibration paper)
- Guo et al. (2017) — *On Calibration of Modern Neural Networks* (temperature scaling and reliability diagrams)